Name:

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Task: Exercise 1 (SQL)

Question 1

(a)	CREATE TABLE Book (
	    book_id INT PRIMARY KEY,
	    title VARCHAR(100) NOT NULL,
	    genre VARCHAR(100),
	    authorID INT,
	    FOREIGN KEY(authorID) REFERENCES Author(authorID)
	);
	
	CREATE TABLE Loan (
	    loan_id VARCHAR(10) PRIMARY KEY,
	    bookID INT,
	    FOREIGN KEY(book_ID) REFERENCES Books(book_id),
	    member_name VARCHAR(100) NOT NULL,
	    loan_date DATE,
	    due_date DATE,
	    return_date DATE

(b)	ALTER TABLE Book RENAME COLUMN authorID to author_ID;
(c)	SELECT book_id, title FROM Book WHERE genre = ‘Fiction’;
(d)	SELECT * FROM Book WHERE title LIKE '%on%';

(e)	SELECT genre, COUNT(*) AS number_of_books FROM Book, GROUP BY genre;
(f)	UPDATE Book SET Title = ‘Sapiens’ , WHERE book_id = 102;
(g)	DELETE FROM Loan WHERE return_date IS NOT NULL
(h)	INSERT INTO Author (author_id, name) VALUES(4, Brendan Chia)
(i)	SELECT book_id, title, COALESCE(genre, 'Unknown') AS genre, author_id FROM Book;
(j)	SELECT * FROM Loan WHERE due_date BETWEEN ‘2025-09-15’ AND ‘2025-09-19’
(k)	SELECT b.book_id, b.title FROM Book b LEFT JOIN Loan l ON b.book_id = l.book_id WHERE l.loan_id IS NULL;

Question 2

(a)	ALTER TABLE MenuItem ADD CONSTRAINT uniq_item_name(rest_id, item_name)
(b)	ALTER TABLE MenuItem ADD CONSTRAINT check_price CHECK (price > 0) ALTER TABLE OrderItem ADD CONSTRAINT check_qty CHECK (qty > 0)
(c)	SELECT * FROM MenuItem WHERE item_name LIKE ‘%Laksa%’;
(d)	UPDATE MenuItem SET price = price + 0.5 WHERE category = ‘Drink’;
(e)	ALTER TABLE Orders MODIFY COLUMN status VARCHAR(20) CHECK (status IN ('Preparing', 'Delivered', 'Cancelled'));

(f)	WITH RankedItems AS (
	    SELECT
	        rest_id,
	        item_name,
	        price,
	        RANK() OVER(PARTITION BY rest_id ORDER BY price DESC) as price_rank
	    FROM
	        MenuItem
	)
	SELECT
	    rest_id,
	    item_name,
	    price
	FROM
	    RankedItems

	WHERE price_rank = 1;
(g)	SELECT mi.item_id, mi.item_name FROM MenuItem mi LEFT JOIN OrderItem oi ON mi.item_id = oi.item_id WHERE oi.order_id IS NULL;
(h)	SELECT r.rest_name, COUNT(o.order_id) AS delivered_orders_count FROM Restaurant r JOIN Orders o ON r.rest_id = o.rest_id WHERE o.status = 'Delivered' GROUP BY r.rest_name;
(i)	SELECT r.rest_name, COUNT(mi.item_id) AS menu_item_count FROM Restaurant r JOIN MenuItem mi ON r.rest_id = mi.rest_id GROUP BY r.rest_name;
(j)	SELECT item_id, item_name, price FROM MenuItem WHERE category = 'Main' ORDER BY price DESC LIMIT 5;
(k)	DELETE FROM OrderItem WHERE qty = 0;

Question 3

(a)	CREATE TABLE Enroll (
	    Sid VARCHAR(10),
	    Cid VARCHAR(10),
	    Semester VARCHAR(10),
	    Grade VARCHAR(2),
	    PRIMARY KEY (Sid, Cid, Semester),
	    FOREIGN KEY (Sid) REFERENCES Student(Sid),
	    FOREIGN KEY (Cid) REFERENCES Course(Cid)
	);

(b)	ALTER TABLE Student ADD COLUMN GPA DECIMAL(3, 2);
(c)	ALTER TABLE Student ADD CONSTRAINT uq_program_sname UNIQUE (Program, Sname);
(d)	UPDATE Student SET Program = 'SECPH' WHERE Sid = 'AC103';
(e)	INSERT INTO Student (Sid, Sname, Program) VALUES ('AC105', 'Siti Nurhaliza', 'SECRH'); INSERT INTO Enroll (Sid, Cid, Semester, Grade) VALUES ('AC105', 'SECP3623', '202520261', NULL); INSERT INTO Enroll (Sid, Cid, Semester, Grade) VALUES ('AC105', 'SECR2414', '202520261', NULL);
(f)	SELECT c.Cid, c.Cname FROM Course c LEFT JOIN Enroll e ON c.Cid = e.Cid WHERE e.Sid IS NULL;
(g)	SELECT * FROM Enroll WHERE Semester = '202520261' AND Grade = 'A';
(h)	SELECT s.Sid, s.Sname FROM Student s JOIN Enroll e ON s.Sid = e.Sid

	JOIN Course c ON e.Cid = c.Cid WHERE c.Cname IN ('Database Programming', 'Database') GROUP BY s.Sid, s.Sname HAVING COUNT(DISTINCT c.Cname) = 2;
(i)	SELECT s.Sid, s.Sname FROM Student s LEFT JOIN Enroll e ON s.Sid = e.Sid WHERE e.Cid IS NULL;
(j)	SELECT s.Sid, s.Sname, SUM(c.Credit) AS TotalCredits FROM Student s JOIN Enroll e ON s.Sid = e.Sid JOIN Course c ON e.Cid = c.Cid WHERE e.Semester = '202520261' GROUP BY s.Sid, s.Sname;
(k)	DELETE FROM Enroll WHERE Sid = 'AC104';